

Conjugated Systems & Aromaticity

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July 2026

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1 Diel Alder Reactions

Core Concept 1.1: Cycloadditions

Cycloadditions refer to the reaction of two separate π systems to form a new ring, an interaction that involves neither a nucleophile nor an electrophile.

Cycloadditions have already been encountered in two reactions within Alkenes & Alkynes.

(a) **Ozonolysis** (Core Concept 1.18)

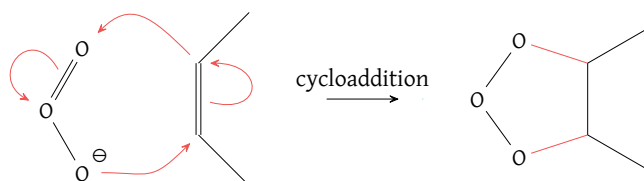


Figure 1: Ozonolysis

(b) **OsO₄ Syn-Dihydroxylation** (Core Concept 1.17)

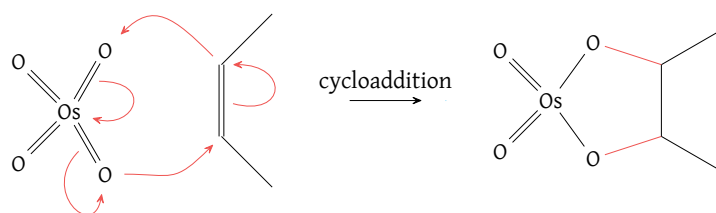


Figure 2: OsO₄ syn-dihydroxylation

Core Concept 1.2: Diel Alder Reactions

Diels-Alder reactions are a type of cycloaddition that combine a diene possessing a minimum of $4\pi e^-$ s with a dienophile possessing a minimum of $2\pi e^-$ s, forming a cyclohexene product.

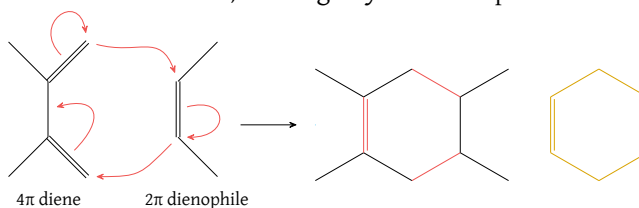


Figure 3: Diel alder reaction; combination of a minimum 4π diene wiht a minimum 2π dienophile

Consequently, diene and dienophile pairings possessing conjugated π systems larger than these stated minimums of 4π and 2π , respectively, can give rise to a substantial diversity of stereoisomeric products.

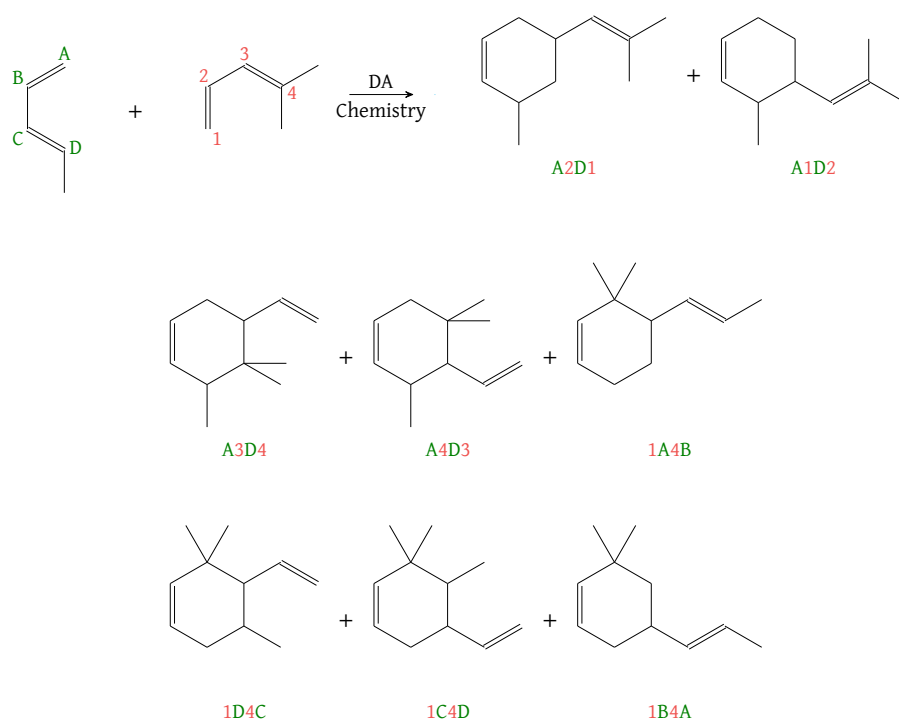
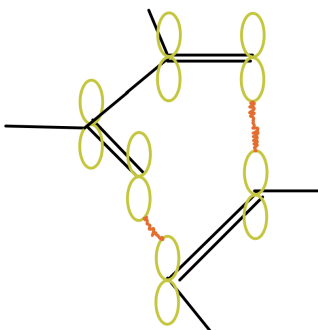


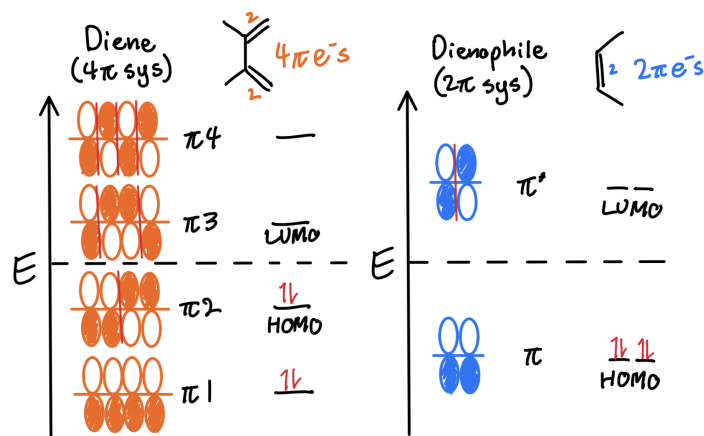
Figure 4: Diels-Alder reaction product diversity

Core Concept 1.3: Orbitals of Diels-Alder Reactions

In Diels-Alder reactions, the π systems of the diene and dienophile approach one another in a vertical orientation, enabling them to generate new σ overlaps.

Figure 5: Diene and dienophile vertical interaction; *p orbitals overlap to form σ interactions*

Consider the energy diagrams of the respective conjugated π systems of the diene and dienophile.

Figure 6: Conjugated π system energy diagrams of a diene and dienophile

A key observation follows from examining which energy levels interact with which: the HOMO of one reagent interacts with the LUMO of the other.

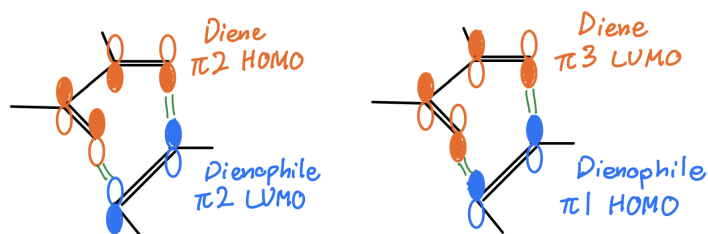


Figure 7: HOMO-LUMO interaction versions between diel alder reagents

Core Concept 1.4: Diel Alder Stereochemistry

Diels-Alder reactions can further be categorized according to the distinct stereochemical approaches the reagents may adopt.

- (a) **Top** and **bottom** Diels-Alder reactions refer, respectively, to the dienophile approaching the diene from **above** or **below** its plane.



Figure 8: Bottom (left) and top (right) attacks by the dienophile in a diel alder reaction

- (b) **Endo** and **exo** distinctions refer to whether the substituents of the dienophile are oriented **towards** or **away from** the corresponding diene.

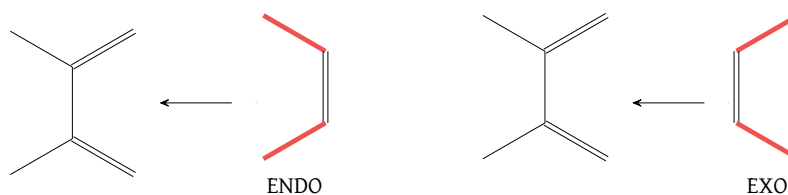


Figure 9: Endo (*left*) and exo (*right*) attack by the dienophile in a Diels-Alder reaction

It is important to note that the bonds oriented towards the diene are flipped about the dienophile's double bond, in the direction opposite to the diene's approach.

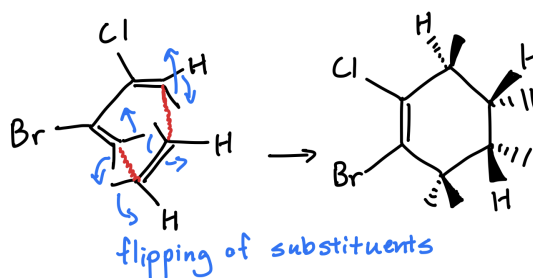


Figure 10: Flipping of dienophile substituents pointing towards the diene during a Diels-Alder reaction

In general, this flipping process can be summarized by the following outline reaction.

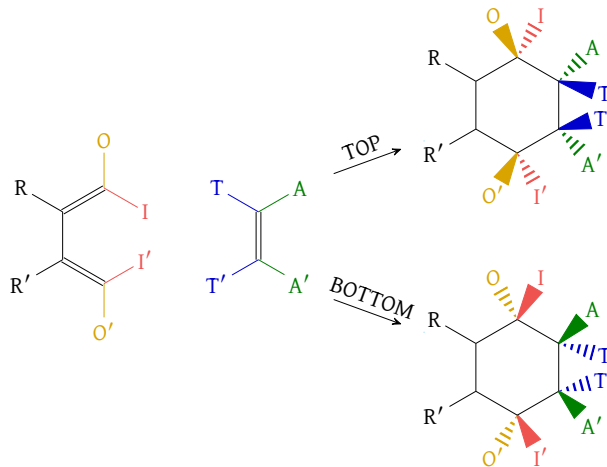


Figure 11: Diastereoselective outcomes resulting from top-face (*top*) versus bottom-face (*bottom*) attack during a cycloaddition reaction

As practice, consider the following Diels-Alder reaction, in which the objective is to predict all non-dimeric Diels-Alder products.

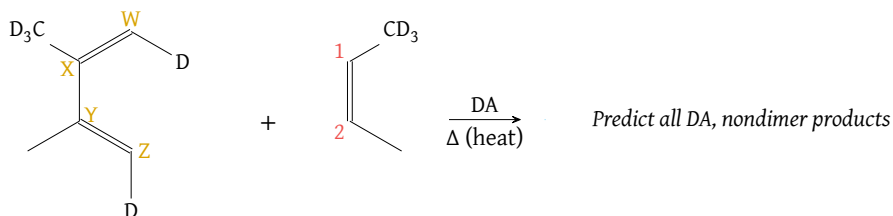


Figure 12: Example diel alder reaction

Two possible constitutional isomers can form, corresponding to the two distinct positionings the dienophile may adopt.

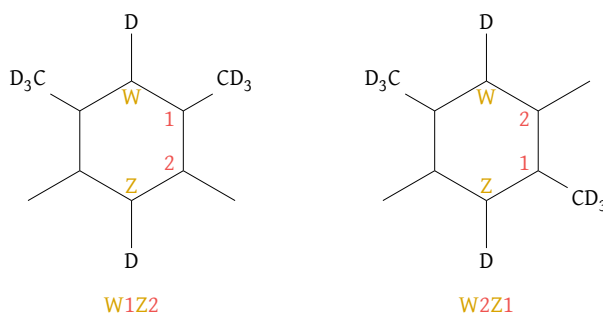


Figure 13: Isomer product possibilities of the reaction shown in Figure 12

It is also important to note that for each of these diene–dienophile arrangements, four distinct stereochemical approaches exist—endo top, endo bottom, exo top, and exo bottom—consistent with the general outline given in Figure 11.

1.1 Rates of Diel Alder Reactions

Core Concept 1.5: Diene Orientation

Diels alder reactions are more favorable when the resulting ring has minimal strain. The **s-cis** diene orientation does this better than the **s-trans** orientation as shown below. The *s* in *s-cis* and *s-trans* stands for *specifically along single bond* as clarification because conventionally *cis* and *trans* in the context of alkenes look at the double bond as the reference bond.

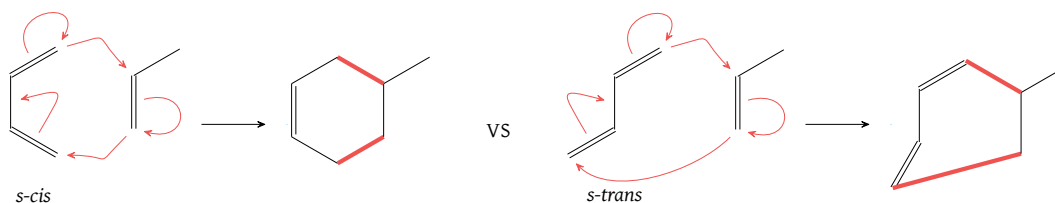


Figure 14: *s-cis* (left) vs *s-trans* (right) diene orientation; *s-trans* orientation induces aggressive ring strain; a ring needs to be at least 8 atoms large to accommodate a *trans* π bond, thus the above *s-trans* reaction will not proceed

Take the following dienes for instance.

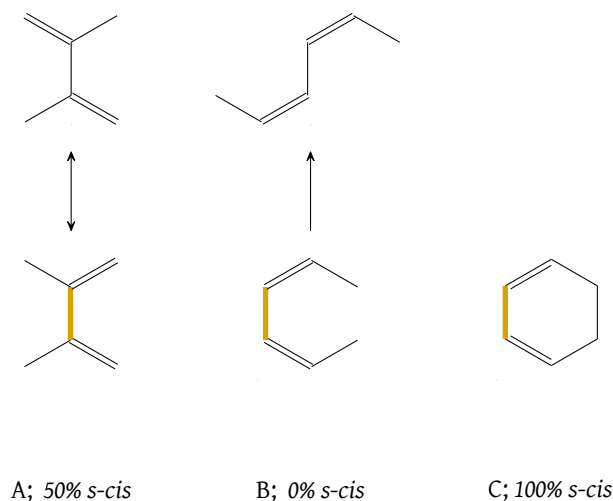


Figure 15: Differences in s-cis orientation dominance across different dienes; *Diels-Alder reaction rates favor diene C the most and diene B the least*

The relative diels alder reaction rates is $C > A > B$. C reigns superior because its structure restricts rotation along the relevant single bond, preventing any rearrangement to an s-trans arrangement. A and B, on the other hand, both can rotate along the relevant single bond, however B favors the s-trans arrangement because the s-cis arrangement has a comparatively higher level of steric hinderance between the alkenyl substituents to the relevant single bond.

Core Concept 1.6: Secondary Orbital Overlap

Under certain scenarios, orbitals not directly responsible for new bond formation between the diene and dienophile can undergo a secondary overlap to lower the overall E_a . Take a look at the following diels alder interaction between two cyclopentenes.

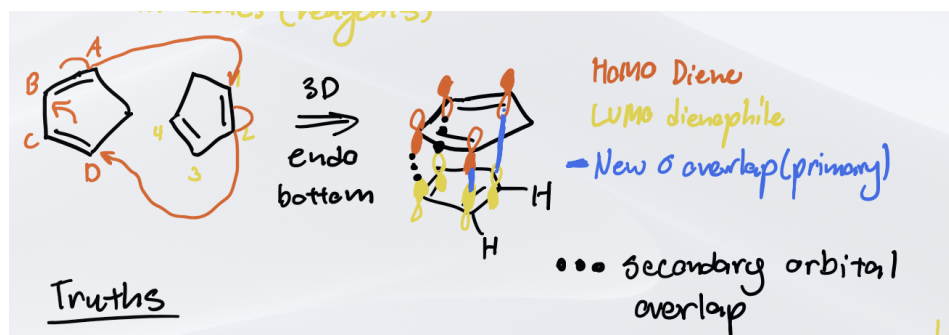


Figure 16

Figure 17: caption

It's important to keep in mind that although favorable mixing of secondary orbital overlaps in the transition state will help stabilize that transition state, sterics still does not favor the endo approach. As a result,

ambiguous diene dienophile pairs will usually need to undergo experiments to see which effect is more important in each case.

Core Concept 1.7: Electronic Effects

Diels Alder reactions progress faster when the energies of the interacting orbitals are closer to one another. In other words, the smaller the HOMO-LUMO gap between reacting molecules, the faster the rate. This usually occurs when more conjugation occurs.

However, while more conjugation typically shrinks the HOMO-LUMO gap and speeds up a reaction, it also alters the wave topology (*the layout of the nodes*). If the orbital with the ideal energy gap happens to place a node on an essential bonding atom, that orbital becomes useless for the reaction, forcing the system to use a higher-energy, slower pathway. Symmetry and overlap must always be satisfied before energy gaps even matter.

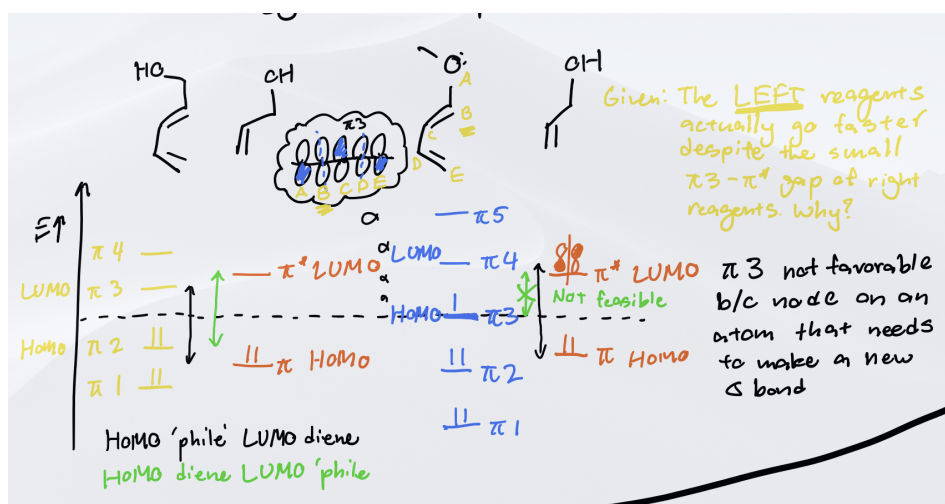


Figure 18: The left reagents actually go faster despite the small $\pi_3 - \pi^*$ gap of the right reagents because π_3 is not favorable because there is a node of the atom that needs to make a new σ bond

Core Concept 1.8: Groundstate Destabilization

Given that two sets of reagents that are similar will have similar transition states leading to the **same** product, the reactant pair in which the ground states of the initial diene and dienophile start at a higher energy levels (*due to something such as steric hinderance*) will have a faster diels alder reaction rate. Take the following diene dineophile pairs leading to the same product for instance.

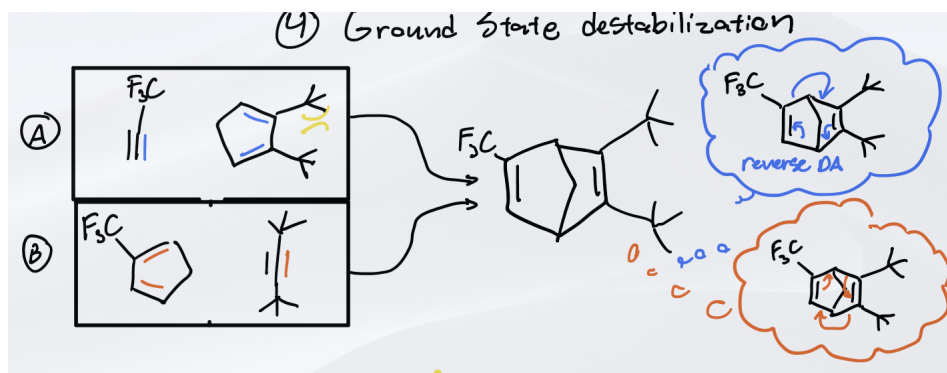


Figure 19

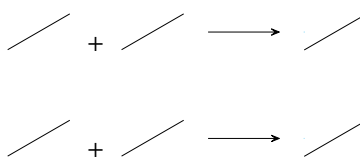


Figure 20: caption

The energy level for the right diene-dienophile pair possesses a greater level of steric hindrance between the isobutyl substituents within the diene, placing it at a higher energy level relative to the left diene-dienophile pair.

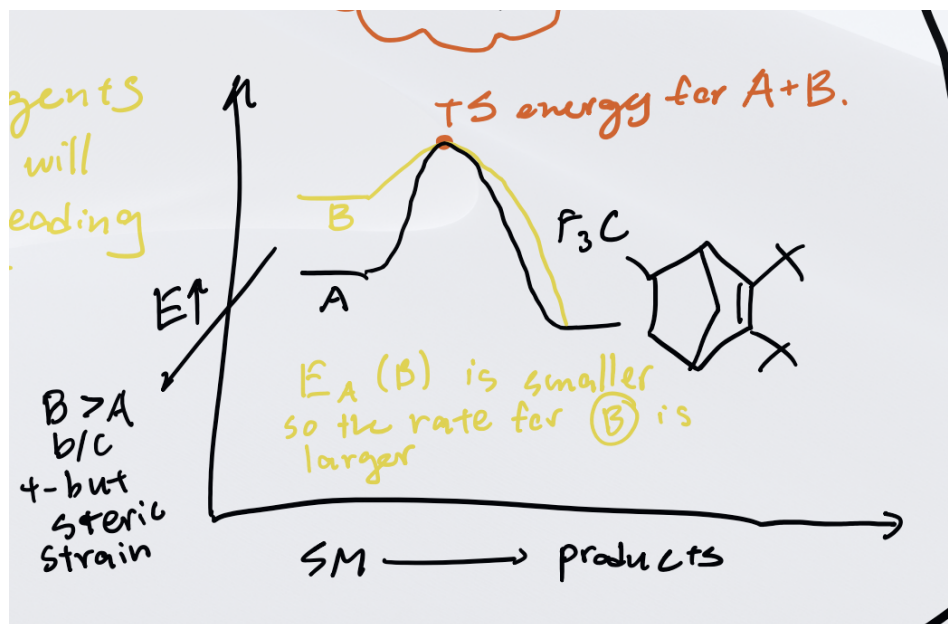


Figure 21

As shown in the energy diagram above, the E_a is much lower for the more unstable reactant pair and thus progresses at a much faster rate because the total ΔH is lower.